# Practical 2
## DC Motor Speed Control with P and PI

Name / group: _____ Date: _____

**Main learning task:** explain why proportional control leaves a speed offset and how integral action removes it. Reuse the Practical 1 model, simulate P and PI, test a reviewed PI candidate, then verify three fresh repeats. Position is optional in simulation only.

Website: **Open Practical 2 – speed control with P and PI**
https://ericwangjcu.github.io/EE3600-Motor-Practicals/practical2-control.html

### 1. Carry forward the Practical 1 model

Select the same motor used in Practical 1. The two website choices are:

| Motor selection | Gearbox ratio, $N$ | Motor encoder counts/rev |
|---|---|---|
| 6 V / 280 RPM output | 21.3:1 | 44 |
| 6 V / 1360 RPM output | 4.4:1 | 44 |

Both use the same P2 UF2. The selector changes the gearbox conversion, not your measured model or controller gains.

Use your existing experimental gain $\overline{K}$, time constant $\overline{\tau}$ and gearbox ratio $N$. Do not repeat identification, voltage/speed measurements, physical-parameter derivations or P1 model validation.

$$G_{P1}(s) = \frac{\overline{K}}{\overline{\tau}s + 1} \qquad \text{[gearbox-output rad/s per volt].}$$

The website uses **motor-shaft RPM** and **fractional duty** $u$ ($0.40 = 40\%$). With the actual declared supply $V_s$ and nominal approximation $V_{\text{motor}} \approx V_s u$:

$$G_n(s) = \frac{K}{\tau s + 1}, \qquad K = \overline{K} N \frac{60}{2\pi} V_s, \qquad \tau = \overline{\tau}.$$

$K$: motor RPM per duty fraction. $\tau$: seconds.

The conversion changes the gain, not the time constant. $K$ is the plant gain, not controller gain $K_p$. Keep your own motor/gearbox/load and record the supply and units. Driver losses and nonlinear behaviour are not separately represented by this voltage scaling.

| $\overline{K}$ [rad/s/V] | $\tau$ [s] | $N$ | $V_s$ [V] | Converted $K$ |
|---|---|---|---|---|
| | | | | |

**Website section 1:** select your motor and actual supply, enter your P1 gain and time constant, then choose the agreed target in motor RPM. Select **Use my P1 model**. The model is converted once; no controller gains are applied.

Follow the website in order: 1 P1 model → 2 P only → 3 Add I → 4 Targets → 5 Pico test → 6 Results. Use **Next/Back** or the numbered sections. **Before the Pico test, complete the new hardware and startup instructions in Sections 5A–5B. P2 motor-power wiring is different from P1.**

## 2. Analyse and simulate P-only control

Use unity negative feedback, $e = r - n$, with $C(s) = K_p$. Derive the closed-loop response:

$$T_P(s) = \frac{KK_p}{\tau s + 1 + KK_p}, \qquad \tau_{\text{cl}} = \frac{\tau}{1 + KK_p},$$

$$n_\infty = \frac{rKK_p}{1 + KK_p}, \qquad \frac{e_{\text{ss}}}{r} = \frac{1}{1 + KK_p}.$$

**Choose an initial P gain**

Use the converted plant gain $K$ from section 1. Choose a modest initial loop gain:

$$KK_{p,0} = 0.5, \qquad K_{p,0} = \frac{0.5}{K}, \qquad n_\infty = \frac{r}{3}, \qquad \tau_{\text{cl}} = \frac{\tau}{1.5}.$$

This deliberately demonstrates offset. The factor 0.5 is a teaching starting choice, not an optimal-gain formula. Gain units are duty/RPM.

1. Set $K_i = K_d = 0$. Enter $K_{p,0}$ as the base P gain. Compare $0.5\times$, $1\times$ and $2\times$ this base at the same reference.
2. Plot reference and speed; record final speed, offset, time constant and required duty.
3. Explain why P needs a remaining error to supply the drive needed for steady rotation.

For this ideal first-order model, positive P gives a monotonic response with **no overshoot**. Increasing P speeds the response and reduces, but does not eliminate, the offset. Settling around the wrong final speed is not successful target tracking. **A physical P-only trial is not required.**

The control website's **Compare P** panel uses the ideal model. The separate legacy limited simulator includes a fixed actuator lag, so its P response need not have the same no-overshoot property.

## 3. Add I and tune PI in simulation

$$C_{PI}(s) = K_p + \frac{K_i}{s}, \qquad T_{PI}(s) = \frac{K(K_p s + K_i)}{\tau s^2 + (1 + KK_p)s + KK_i}.$$

I accumulates persistent error. At zero error, the accumulated integral contribution can remain nonzero and supply the required steady drive. Zero ideal step error assumes a stable loop and an achievable target.
**Choose an initial I gain, then refine**

Keep your selected P gain fixed and D = 0. Express PI using the integral time:

$$C(s) = K_p \left(1 + \frac{1}{T_i s}\right), \qquad K_i = \frac{K_p}{T_i}.$$

$T_i$ is the integral time in seconds, not the settling time. For constant error, the integral contribution equals the proportional contribution after this time.

$$T_{i,0} = 5\tau, \qquad K_{i,0} = \frac{K_p}{5\tau} \quad [\text{duty}/(\text{RPM}\cdot\text{s})].$$

The factor 5 starts integral action slowly relative to the plant. It is a simulation heuristic, not a lecture-derived optimum or an approved hardware setting.

1. Enter $K_{i,0}$ as the base I gain and compare $0.5\times$, $1\times$ and $2\times$ it, holding P fixed. Inspect speed, duty, overshoot, rise, settling and final error.
2. If tracking is too slow, increase I in simulation. If overshoot or oscillation grows, reduce I. Check the whole trace; a smaller final error alone is not a pass.
3. If I changes alone cannot meet the targets, adjust P one step at a time and repeat the I comparison. Record each candidate, its results and the reason for the change; save the initial and selected designs.
4. Check the complete response, including the PI zero at $-K_i/K_p$. Denominator-only overshoot formulas do not determine this response.
5. Before hardware, check duty limits, the 10 ms controller interval and declared filtering in MATLAB/Simulink. Do not add measurement lag twice if included in the P1 fit. Obtain tutor review before entering hardware gains.

**Worked simulation start only:** if $K = 5955.553$ and $\tau = 0.051$ s, $K_{p,0} = 0.00008396$, $T_{i,0} = 0.255$ s and $K_{i,0} = 0.0003292$. Use your own $K$ and $\tau$. If you select a different P, recalculate the starting I from that P. These are not the final bench gains.

**PI is sufficient when it meets the requirements. D was not needed in the reported successful speed tests.** No extra time-constant helper or pole-choice exercise is required.

**How the three lecture methods apply**

| Method | Use in this practical |
|---|---|
| Model-based analysis | Derive P/PI behaviour and explain gain effects. |
| Computational tuning | Adjust or search gains in simulation. State the settings and criteria, then validate one reviewed candidate. |
| Ziegler–Nichols | Explain both methods and assess applicability. The ideal model has no positive reaction-curve delay and no finite positive ultimate gain. Use a separate suitable tutor example for numerical practice. |

**These are not three compulsory hardware exercises.** Do not invent a delay or ultimate gain, or increase physical gain to force sustained oscillation.

## 4. Speed requirements and measurement conventions

Use one agreed achievable positive step from rest, with the same target, supply, load and limits throughout. The reported reference kit used 1000 motor RPM at 6 V; this is not a universal target for every kit.

| Quantity | Requirement |
|---|---|
| Overshoot | $\leq$ 5% of the commanded step, beyond the final target. |
| 10–90% rise | $\leq$ 0.50 s. Also report command-to-10% time to expose startup delay. |
| Settling | $\leq$ 1.00 s from the command event; all later assessment samples in band, with $\geq$ 2 s final in-band dwell. |
| Final error | Last-2-second mean absolute error $\leq \max(2\%\text{ of target}, 2q)$. Report signed bias too. |

**Encoder-aware assessment band**

For $C_m$ decoded counts per motor revolution and the 50 ms speed-measurement window:

$$q = \frac{60}{0.050 C_m} \quad \text{motor RPM.}$$

The settling band is centred on the commanded final target $r_f$:

$$r_f \pm \max\left(0.02|r_f - r_0|,\ 2q\right).$$

For $C_m = 44$, $2q = 54.545$ motor RPM. At a 1000 RPM target this is $\pm 5.45\%$, **not strict** $\pm 2\%$. State which term governs the band. The ideal simulation comparison reports strict 2% separately.

**Assess against the commanded target.** Measure time from the command, not from the start of motion. Faults, incomplete recordings or insufficient final dwell cannot pass.

| Agreed target: motor RPM | Assessment-band half-width: motor RPM |
|---|---|

## 5A. Hardware connections – what changes from P1

**Practical 1:** the bench supply drove the motor directly; the Pico measured the encoder.
**Practical 2:** the bench supply powers the **motor-driver board**, and the motor connects to **Motor 1**. The Pico now controls the drive as well as measuring speed. The encoder wiring stays the same.

**Make all connections with power removed**

Switch **OFF** the bench supply and the driver-board switch, and unplug the Pico USB cable before moving wires. Secure the motor with the shaft clear.

1. **Fit/check the Pico.** Use the original RP2040 Pico seated in the Kitronik 5331 board's two header sockets. Match the Pico outline/orientation on the board; check for reversed or offset pins. Do not remove an already-correct fitted Pico.
2. **Remove the direct P1 motor-supply connection.** Disconnect **both motor winding leads** from the bench supply. Move the motor's **red and white leads** to the **two Motor 1 output terminals**, one lead in each terminal. Use the tutor-marked lead order; leave Motor 2 unused.
3. **Power the board, not the motor directly.** Connect bench supply **+ to the board POWER +** terminal and bench supply **– to POWER – / GND**, following the board markings. For the two 6 V kits, use a **6 V supply** unless the tutor explicitly approves a different configuration. Leave the supply and board switch OFF for now.
4. **Retain/check the encoder wiring** using the table below. Keep the encoder's low-voltage wiring separate from the motor-power terminals. If wire colours differ, identify the signals with the tutor rather than guessing.

| Wire on the class kit | Pico connection | Purpose |
|---|---|---|
| Yellow | GP26 | Encoder A signal |
| Green | GP27 | Encoder B signal |
| Blue | 3V3 (OUT), or board 3V breakout | Encoder supply; not motor supply |
| Black | GND | Encoder ground, common with Pico/board ground |

GP26 and GP27 are GPIO names, not physical pin numbers. Retain the existing P1 harness; do not substitute GP28 for GP26. The board's header sockets already carry the Motor 1 control signals GP2/GP3 and common ground; no extra motor-control jumpers are required.

**Do not leave the bench supply connected directly to the motor.** Do not connect either Motor 1 output to supply + or GND, and do not power the motor from Pico 3V3 or a GPIO. The encoder must not receive the 6 V motor supply. USB alone is not the motor-power source for this practical.

**Connection check:** bench supply → board POWER input; Motor 1 → red/white motor leads; encoder → GP26/GP27, 3V3 and GND. The PC connects to the **Pico USB port** with a data-capable cable in Section 5B.

Connection basis: P1 task sheet, Section 3 (class-kit colours); P2 hardware guide/firmware (Motor 1 and GP26/GP27); Kitronik 5331 datasheet, pp. 1–2 (board power, sockets and Motor 1).

## 5B. Load P2 firmware and bring the hardware online

### 1. Replace P1 firmware once, if needed

The P1 encoder-only firmware does not run this controller. Use **EE3600_Practical_2_Pico.uf2** for **both motor versions**. If the Pico already reports **EE3600_P2 1.0.0**, skip re-flashing.

Download from the website's **P2 firmware** link, or **open the P2 UF2 download**.

1. Keep the bench motor supply disconnected/OFF and the driver-board switch OFF. Unplug the Pico USB cable.
2. Hold **BOOTSEL** on the Pico while reconnecting its USB cable to the PC. Release BOOTSEL after the **RPI-RP2** drive appears.
3. Copy **EE3600_Practical_2_Pico.uf2** onto RPI-RP2. The drive disappears and the Pico reboots. Do not copy the P1 recovery UF2 or install MicroPython for this task.

### 2. Connect to the website while motor power is OFF

1. Open the Practical 2 website in desktop **Chrome or Edge**. Close Thonny, serial monitors and other pages using this Pico's port.
2. In website section 1 select the **actual motor** (280 RPM / 21.3:1 or 1360 RPM / 4.4:1) and **actual supply voltage**. Complete the model and simulation work in sections 1–4.
3. Go to website section 5. Click **Connect Pico**, select the Pico serial port and grant the browser permission. Confirm compatible P2 firmware and a **DISARMED** state. Connecting must not start the motor.
4. Check **Tutor settings**, including the maximum duty and normal **10 s trial**. Click **Apply configuration** while disarmed, then check the readback: $C_m = 44$, the selected gearbox ratio, actual supply and agreed limits. Do not enter approximately 937 output-shaft counts as motor counts.

### 3. Check, then switch on motor power

1. Ask the tutor to check the wiring, approved supply-current limit, duty ceiling and motor/encoder direction pairing. The board datasheet specifies **1.5 A maximum per motor channel**; a supply marked 3 A is not an instruction to set a 3 A current limit. Do not carry out a stall test.
2. With the shaft clear, enable the bench output and the board switch. Check that the Pico still reports **DISARMED**, with **zero applied duty and zero speed**. If switching power resets the Pico, reconnect and reapply the configuration before continuing.
3. Enter the **reviewed PI candidate**, with $K_d = 0$. Start only by selecting **Arm & run speed trial**. Observe speed and applied duty; keep **STOP** accessible.

For a positive command, motion must give positive speed feedback. If the motor runs but feedback is zero/negative, it runs away, stalls, or a fault appears: select **STOP**, switch off motor power and call the tutor. Do not correct wiring while powered, increase limits, bypass a fault or repeatedly re-arm. STOP removes drive; the shaft can coast.

**At the end:** STOP; wait for DISARMED and coast-down to zero speed; switch the board and bench supply OFF; save the recording, then disconnect the website/USB. Do not touch or rewire a moving motor.

## 5C. Website and Pico workflow

The website sections 1–6 follow this sheet. Only one section is shown at a time; use **Next/Back** or the numbered section buttons. The motor selector is in section 1. **Complete Sections 5A–5B before the physical test.**

1. **Section 1 – P1 model.** Select your motor and actual supply, enter your own $\overline{K}$ and $\tau$, and choose the agreed target in motor RPM. Select **Use my P1 model**. The model is converted once for the following simulation sections; no gains are applied.
2. **Section 2 – P only.** Enter a base P gain and compare $0.5\times$, $1\times$ and $2\times$ P. Save the simulated CSV and record the offset and speed of response.
3. **Section 3 – Add I.** Keep the selected base P fixed; compare three I settings. Refine and justify your candidate. Use MATLAB/Simulink to check the declared limits and sampled behaviour before tutor review.
4. **Section 4 – Targets.** Review the response targets and encoder-aware band in section 4 of this sheet.
5. **Section 5 – Pico test.** Use desktop Chrome/Edge and the P2 UF2. Select **Connect Pico**, then **Apply configuration** while disarmed. The chosen profile sets $N$ and motor counts per revolution automatically; check the displayed readback and tutor-approved duty limit. Enter your reviewed $K_p$ and $K_i$. D remains zero and the normal trial is 10 s.
6. **Arm & run speed trial.** Inspect speed and duty. **STOP** removes drive; allow coast-down. A fault requires tutor diagnosis, not an automatic retry.
7. **Section 6 – Results.** Select **Analyse response** $\rightarrow$ **Save CSV**. Save before another trial. Stop before changing a gain; document each refinement.
8. **Freeze and repeat three times.** Retain every attempt, reload saved P2 CSVs and check reproducible metrics. Do not change gains between final repeats.

**Gain units are the same in the simplified comparison and hardware panels.**
$K_p$: duty fraction/RPM. $K_i$: duty fraction/(RPM·s).
Only the separate legacy simulator uses percent duty/output RPM: divide those gains by $100 \times N$ before hardware entry. Unit conversion alone does not match filtering or sampling.

Do not apply a 67% minimum-duty clamp or hidden starting boost. The report only establishes that 67% produced movement in that short check, not an exact threshold. A 100% ceiling was approved for the reported 6 V bench, not every kit. **On a fault, stop and obtain tutor diagnosis before restarting.** Physical position controls are not part of this practical.

**Reviewed PI candidate:** $K_p =$ _____ $K_i =$ _____ $K_d = 0$.

## 6. Results and submission

| Case | Gains | Over-shoot (%) | Rise (s) | Settling (s) | Final MAE / bias (RPM) |
|---|---|---|---|---|---|
| P simulations | | | | | |
| PI simulations initial/final | | | | | |
| Initial hardware | | | | | |
| Final repeat 1 | | | | | |
| Final repeat 2 | | | | | |
| Final repeat 3 | | | | | |

Submit carried-forward P1 values and the unit conversion, P/PI analysis, simulation/tuning record, new P2 CSVs, speed/reference/duty plots, the comparison table and discussion. **Do not resubmit P1 identification work.** Preserve all raw timestamps; exclude only explicitly reported pre-command transition frames from assessment, never startup delay or failed responses.

Explain P's offset, how I supplies steady drive, what increasing I changed, model/hardware differences, whether each final repeat passed, the assessment band, and why the two Ziegler–Nichols methods do not directly fit this ideal model. **Reported bench reference – not a prescribed student answer**

One unloaded 6 V kit, target 1000 motor RPM, $N = 21.3$ and $C_m = 44$: $K_p = 0.0001284515583$, $K_i = 0.0010074632025$, $K_d = 0$. Three fresh reported repeats met numerical speed targets. Worst overshoot: 3.655%; rise: 0.2833 s; settling: 0.962 s; final MAE: 13.465 RPM; band: $\pm 54.545$ RPM. These are supplied report values, not new website measurements. A later $N = 21.6$ estimate does not rewrite those historical runs.

## 7. Optional position extension – simulation only

$$G_\theta(s) = \frac{6}{N}\frac{G_n(s)}{s} \qquad \text{[output degrees / duty fraction]}.$$

Use the selected lecture's position exercise in MATLAB/Simulink: compare P and PD/PID, state the controller structure and reference, and assess its optional simulation criteria (overshoot $\leq 5\%$, $0.25 \leq$ settling time $\leq 0.80$ s and zero ideal final error). Do not copy speed gains directly. Both reported physical position candidates faulted; neither is a validated bench demonstration. **No position hardware trial is required. Instructor release status**

The supplied bench report supports numerical speed performance only at the stated operating point. The revised page needs a physical native-browser trial, CSV export/re-import against real records and a separate physical USB-removal check before complete end-to-end validation is claimed. Browser tests with synthetic CSVs do not establish hardware performance.

Basis: coordinator's six-page formatted task sheet, existing P1 wiring/model, selected PID lecture and supplied bench reports. This revision adds Sections 5A–5B and their cross-references. Initial P/PI calculations, worked example and assessment targets are retained. Lecture slides, Practical 1 and firmware are unchanged. Board connection details were checked against the Kitronik 5331 datasheet and the P2 project hardware guide.